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The Oosterhoff Dichotomy in Halo Globular Clusters Revisited

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Abstract. Galactic globular clusters divide into two groups by the mean period of their RRab stars: Oosterhoff I (⟨P_{ab}⟩ ≈ 0.55 d) and Oosterhoff II (⟨P_{ab}⟩ ≈ 0.64 d). Using homogeneous data for 61 clusters, we confirm the gap and find that the RRc fraction is higher in Oosterhoff II systems.

1. Sample

We compiled RR Lyrae catalogs for 61 halo clusters with at least 10 RRab stars each.

2. Results

The distribution of ⟨P_{ab}⟩ is bimodal, with few clusters in the 0.58–0.62 d range. Oosterhoff II clusters show RRc fractions of ~0.45 versus ~0.20 for Oosterhoff I.

Table 1. Summary of key values.

Group	⟨P_{ab}⟩ (d)	N_c/N_{tot}
Oo I	0.55	0.20
Oo II	0.64	0.45

References

Okafor, N., & Lindqvist, E. 2023, Journal of Synthetic Astrophysics, 12, 101